VPN IPSec IKEv2 Site-to-Site con Túnel GRE (GRE over IPSec)

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Matrícula: 2025-0773
Práctica: P4

Objetivo

Configurar una VPN Site-to-Site IPSec IKEv2 utilizando un túnel **GRE (Generic Routing Encapsulation)** protegido mediante IPSec en modo Transporte (GRE over IPSec), entre dos routers Cisco CSR1000v, permitiendo la comunicación cifrada entre dos redes LAN a través de un router ISP. Esta práctica combina la arquitectura GRE over IPSec de P3 con el protocolo de negociación IKEv2 de P4/P5, representando la implementación más moderna de esta topología.

La diferencia fundamental respecto a P3 (IKEv1 GRE over IPSec) es la sustitución de `crypto isakmp policy/key` por el conjunto `crypto ikev2 proposal/policy/keyring/profile`, y la adición de `set ikev2-profile` en el crypto map. La arquitectura GRE + IPSec Transport permanece idéntica.

Comparación P3 (IKEv1) vs P6 (IKEv2) — GRE over IPSec

Característica	P3 (IKEv1 GRE)	P6 (IKEv2 GRE)
Negociación fase 1	`crypto isakmp policy`	`crypto ikev2 proposal/policy/profile`
Llave precompartida	`crypto isakmp key ... address`	`crypto ikev2 keyring` (bloque peer)
Referencia en crypto map	Implícita (IKEv1 por defecto)	`set ikev2-profile PROF-P6`
Mensajes de negociación	6-9 (Main Mode)	4 (IKE_SA_INIT + IKE_AUTH)
Comando de verificación	`show crypto isakmp sa`	`show crypto ikev2 sa`
Estado SA activa	`QM_IDLE`	`READY`
Túnel GRE	Igual	Igual
Modo IPSec	Transport	Transport
ACL (tráfico interesante)	`permit gre host X host Y`	`permit gre host X host Y`

Topología

```
[PC1]--[SW-A]--[R1]--------[ISP]--------[R2]--[SW-B]--[PC2]
        LAN-A   Gi3    Gi1       Gi2    Gi2    LAN-B
              25.7.73.169  25.7.73.161  25.7.73.165  25.7.73.166  25.7.73.177

                       Tunnel0 (GRE)         Tunnel0 (GRE)
                  25.7.73.185 <----------> 25.7.73.186
                     Protegido por IPSec IKEv2 (modo Transport)
```

Direccionamiento IP

Dispositivo	**Interfaz**	**Dirección IP**	**Máscara**	**Descripción**
R1	GigabitEthernet1	25.7.73.161	255.255.255.252	WAN R1 → ISP
ISP	GigabitEthernet1	25.7.73.162	255.255.255.252	WAN ISP → R1
ISP	GigabitEthernet2	25.7.73.165	255.255.255.252	WAN ISP → R2
R2	GigabitEthernet2	25.7.73.166	255.255.255.252	WAN R2 → ISP
R1	GigabitEthernet3	25.7.73.169	255.255.255.248	Gateway LAN-A
PC1	eth0	25.7.73.170	255.255.255.248	Host LAN-A
R2	GigabitEthernet3	25.7.73.177	255.255.255.248	Gateway LAN-B
PC2	eth0	25.7.73.178	255.255.255.248	Host LAN-B
R1	Tunnel0 (GRE)	25.7.73.185	255.255.255.252	Extremo GRE R1
R2	Tunnel0 (GRE)	25.7.73.186	255.255.255.252	Extremo GRE R2

Subredes utilizadas (bloque 25.7.73.160/27)

Subred	**Rango utilizable**	**Uso**
25.7.73.160/30	25.7.73.161 – 25.7.73.162	WAN R1 – ISP
25.7.73.164/30	25.7.73.165 – 25.7.73.166	WAN ISP – R2
25.7.73.168/29	25.7.73.169 – 25.7.73.174	LAN-A
25.7.73.176/29	25.7.73.177 – 25.7.73.182	LAN-B
25.7.73.184/30	25.7.73.185 – 25.7.73.186	Túnel GRE

Parámetros de configuración

Túnel GRE

Parámetro	R1	R2
IP Tunnel	25.7.73.185/30	25.7.73.186/30
Tunnel source	Gi1	Gi2
Tunnel destination	25.7.73.166	25.7.73.161
Tunnel mode	gre ip	gre ip

IKEv2 Proposal

Parámetro	Valor
Cifrado	AES-CBC 256
Integridad	SHA-256
Grupo DH	Group 14 (2048-bit)

IKEv2 Keyring

Parámetro	R1	R2
Peer remoto	R2 (25.7.73.166)	R1 (25.7.73.161)
Pre-shared key	Cisco123!	Cisco123!

IKEv2 Profile

Parámetro	R1	R2
Match identity remote	25.7.73.166/32	25.7.73.161/32
Autenticación local	pre-share	pre-share
Autenticación remota	pre-share	pre-share

Fase 2 — IPSec Transform Set

Parámetro	Valor
Cifrado ESP	AES 256
Integridad ESP	SHA-256 HMAC
Modo	**Transport**

Tráfico interesante (ACL) — cifra protocolo GRE

Router	Regla
R1	`permit gre host 25.7.73.161 host 25.7.73.166`
R2	`permit gre host 25.7.73.166 host 25.7.73.161`

Scripts de configuración

ISP

```
enable
configure terminal
hostname ISP

interface GigabitEthernet1
 ip address 25.7.73.162 255.255.255.252
 no shutdown
 exit

interface GigabitEthernet2
 ip address 25.7.73.165 255.255.255.252
 no shutdown
 exit

end
write memory
```

R1

```
enable
configure terminal
hostname R1

interface GigabitEthernet1
 ip address 25.7.73.161 255.255.255.252
 no shutdown
 exit

interface GigabitEthernet3
 ip address 25.7.73.169 255.255.255.248
 no shutdown
 exit

ip route 0.0.0.0 0.0.0.0 25.7.73.162

! --- Túnel GRE ---
interface Tunnel0
 ip address 25.7.73.185 255.255.255.252
 tunnel source GigabitEthernet1
 tunnel destination 25.7.73.166
 tunnel mode gre ip
 no shutdown
 exit

! --- Ruta hacia LAN-B por el túnel GRE ---
```

```
ip route 25.7.73.176 255.255.255.248 25.7.73.186

! --- IKEv2 Proposal ---
crypto ikev2 proposal PROP-P6
 encryption aes-cbc-256
 integrity sha256
 group 14
 exit

! --- IKEv2 Policy ---
crypto ikev2 policy POL-P6
 proposal PROP-P6
 exit

! --- Keyring ---
crypto ikev2 keyring KEY-P6
 peer R2
  address 25.7.73.166
  pre-shared-key Cisco123!
  exit
 exit

! --- IKEv2 Profile ---
crypto ikev2 profile PROF-P6
 match identity remote address 25.7.73.166 255.255.255.255
 authentication local pre-share
 authentication remote pre-share
 keyring local KEY-P6
 exit

! --- Transform Set en modo Transport ---
crypto ipsec transform-set TS-P6 esp-aes 256 esp-sha256-hmac
 mode transport
 exit

! --- ACL: cifra protocolo GRE (47) entre IPs WAN ---
access-list 101 permit gre host 25.7.73.161 host 25.7.73.166

! --- Crypto Map referenciando IKEv2 profile ---
crypto map CMAP-P6 10 ipsec-isakmp
 set peer 25.7.73.166
 set transform-set TS-P6
 set ikev2-profile PROF-P6
 match address 101
 exit

interface GigabitEthernet1
 crypto map CMAP-P6
 exit

end
write memory
```

R2

```
enable
configure terminal
hostname R2

interface GigabitEthernet2
 ip address 25.7.73.166 255.255.255.252
 no shutdown
 exit

interface GigabitEthernet3
 ip address 25.7.73.177 255.255.255.248
 no shutdown
 exit

ip route 0.0.0.0 0.0.0.0 25.7.73.165

! --- Túnel GRE ---
interface Tunnel0
 ip address 25.7.73.186 255.255.255.252
 tunnel source GigabitEthernet2
 tunnel destination 25.7.73.161
 tunnel mode gre ip
 no shutdown
 exit

! --- Ruta hacia LAN-A por el túnel GRE ---
ip route 25.7.73.168 255.255.255.248 25.7.73.185

! --- IKEv2 Proposal ---
crypto ikev2 proposal PROP-P6
 encryption aes-cbc-256
 integrity sha256
 group 14
 exit

! --- IKEv2 Policy ---
crypto ikev2 policy POL-P6
 proposal PROP-P6
 exit

! --- Keyring ---
crypto ikev2 keyring KEY-P6
 peer R1
  address 25.7.73.161
  pre-shared-key Cisco123!
  exit
 exit

! --- IKEv2 Profile ---
crypto ikev2 profile PROF-P6
```

```
 match identity remote address 25.7.73.161 255.255.255.255
 authentication local pre-share
 authentication remote pre-share
 keyring local KEY-P6
 exit

! --- Transform Set en modo Transport ---
crypto ipsec transform-set TS-P6 esp-aes 256 esp-sha256-hmac
 mode transport
 exit

! --- ACL: cifra protocolo GRE (47) entre IPs WAN ---
access-list 101 permit gre host 25.7.73.166 host 25.7.73.161

! --- Crypto Map referenciando IKEv2 profile ---
crypto map CMAP-P6 10 ipsec-isakmp
 set peer 25.7.73.161
 set transform-set TS-P6
 set ikev2-profile PROF-P6
 match address 101
 exit

interface GigabitEthernet2
 crypto map CMAP-P6
 exit

end
write memory
```

SW-A y SW-B (IOSvL2)

```
enable
configure terminal

interface GigabitEthernet0/0
 switchport mode access
 switchport access vlan 1
 no shutdown
 exit

interface GigabitEthernet0/1
 switchport mode access
 switchport access vlan 1
 no shutdown
 exit

end
write memory
```

PC1 (VPCS)

```
ip 25.7.73.170 255.255.255.248 25.7.73.169
save
```

PC2 (VPCS)

```
ip 25.7.73.178 255.255.255.248 25.7.73.177
save
```

Verificación y pruebas

> Comandos ejecutados en R1 y R2. El ISP no participa de la VPN; los PCs solo ejecutan `ping`.

Comandos de verificación

```
show interface Tunnel0
show crypto ikev2 sa
show crypto ipsec sa
show ip route
```

Resultados obtenidos — R1

show interface Tunnel0:

```
Tunnel0 is up, line protocol is up
  Internet address is 25.7.73.185/30
  Tunnel protocol/transport GRE/IP
  Tunnel source 25.7.73.161 (GigabitEthernet1), destination 25.7.73.166
```

show crypto ikev2 sa:

```
Tunnel-id Local                 Remote                fvrf/ivrf            Status
1         25.7.73.161/500       25.7.73.166/500       none/none            READY
      Encr: AES-CBC, keysize: 256, PRF: SHA256, Hash: SHA256, DH Grp:14, Auth
sign: PSK, Auth verify: PSK
      Life/Active Time: 86400/60 sec
```

show crypto ipsec sa (fragmento):

```
local  ident (addr/mask/prot/port): (25.7.73.161/255.255.255.255/47/0)
remote ident (addr/mask/prot/port): (25.7.73.166/255.255.255.255/47/0)
```

```
#pkts encaps: 5, #pkts encrypt: 5, #pkts digest: 5
#pkts decaps: 4, #pkts decrypt: 4, #pkts verify: 4
in use settings ={Transport, }
Status: ACTIVE(ACTIVE)
```

> Los identificadores muestran protocolo 47 (GRE) — confirma que IPSec está cifrando los paquetes GRE entre las IPs WAN, no el tráfico IP de las LANs directamente.

Pruebas de conectividad

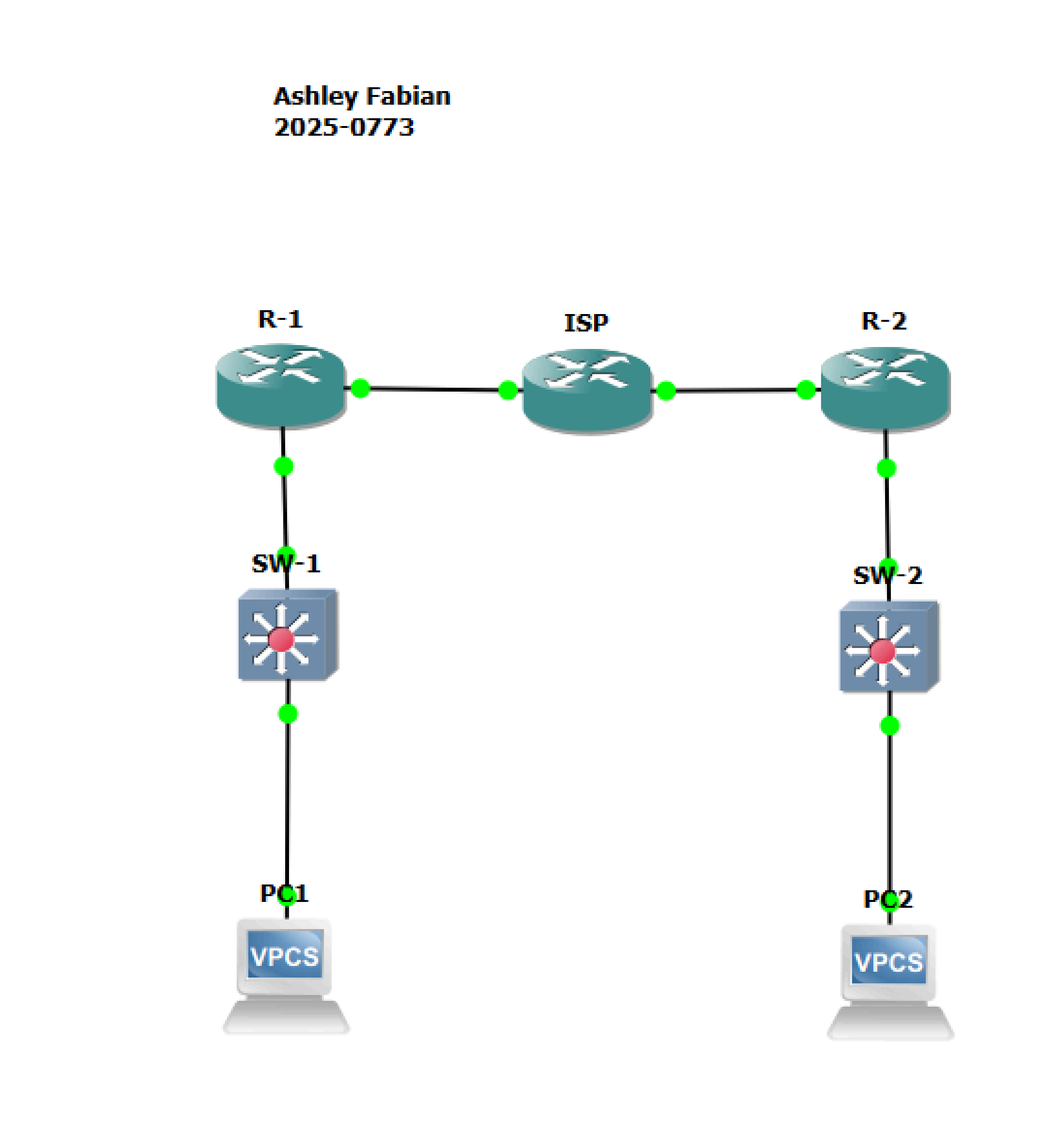
Ping PC1 → PC2:

```
PC1> ping 25.7.73.178
25.7.73.178 icmp_seq=1 timeout
84 bytes from 25.7.73.178 icmp_seq=2 ttl=62 time=474.596 ms
84 bytes from 25.7.73.178 icmp_seq=3 ttl=62 time=55.298 ms
84 bytes from 25.7.73.178 icmp_seq=4 ttl=62 time=105.548 ms
84 bytes from 25.7.73.178 icmp_seq=5 ttl=62 time=47.365 ms
```

> El primer paquete dio timeout durante la negociación IKEv2 (IKE_SA_INIT + IKE_AUTH) y el establecimiento del túnel GRE. Los paquetes subsiguientes viajan cifrados exitosamente. Este comportamiento es idéntico al observado en P3 con IKEv1.

Capturas de pantalla

> Insertar aquí las capturas de:

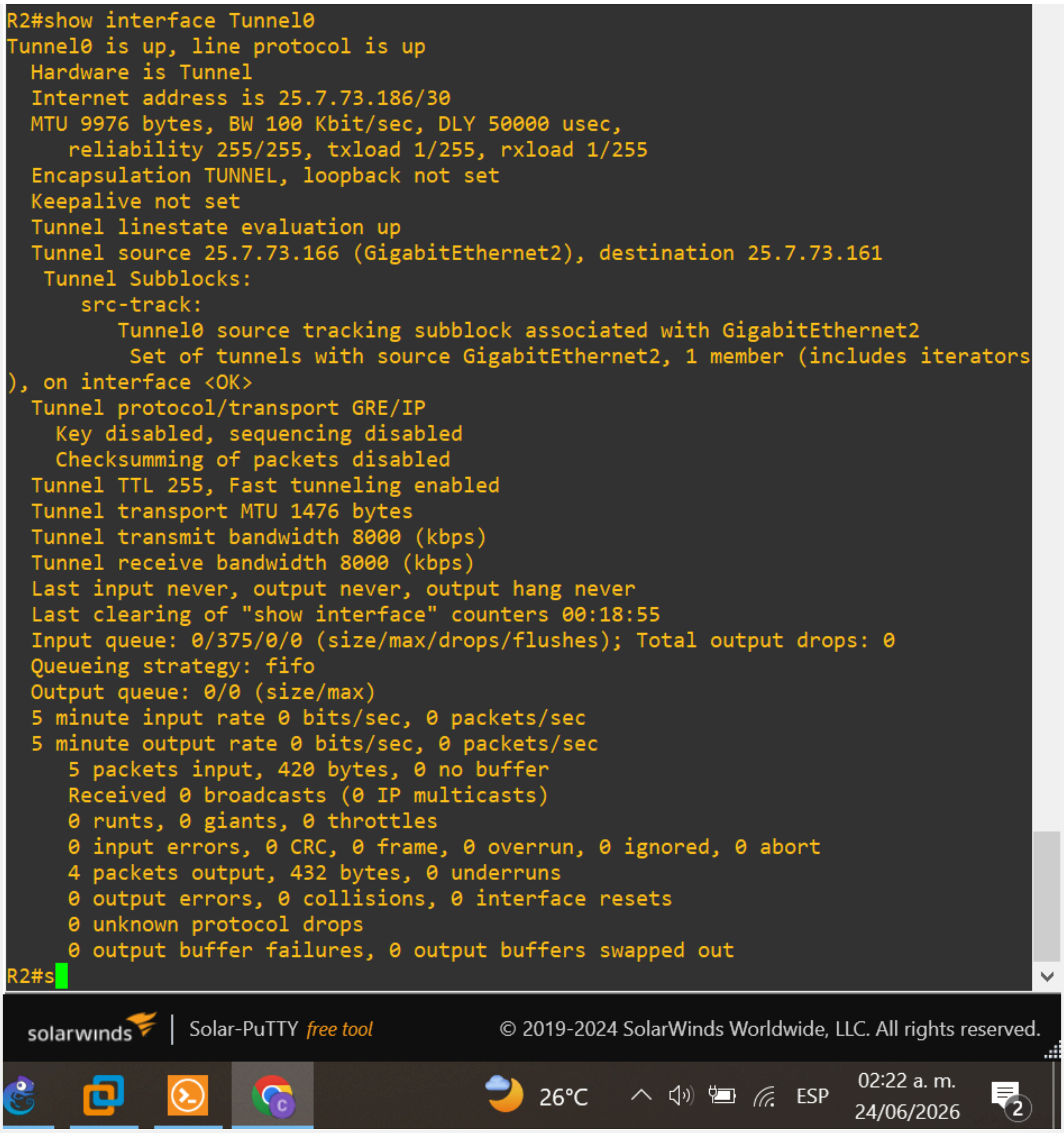
> 1. Topología completa en GNS3 con nombre y matrícula visible
>
> 
>

2. `show interface Tunnel0` en R1 y R2 (GRE/IP, up/up)

```
R1#show interface Tunnel0
Tunnel0 is up, line protocol is up
  Hardware is Tunnel
  Internet address is 25.7.73.185/30
  MTU 9976 bytes, BW 100 Kbit/sec, DLY 50000 usec,
     reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation TUNNEL, loopback not set
  Keepalive not set
  Tunnel linestate evaluation up
  Tunnel source 25.7.73.161 (GigabitEthernet1), destination 25.7.73.166
   Tunnel Subblocks:
      src-track:
         Tunnel0 source tracking subblock associated with GigabitEthernet1
          Set of tunnels with source GigabitEthernet1, 1 member (includes iterators
), on interface <OK>
  Tunnel protocol/transport GRE/IP
    Key disabled, sequencing disabled
    Checksumming of packets disabled
  Tunnel TTL 255, Fast tunneling enabled
  Tunnel transport MTU 1476 bytes
  Tunnel transmit bandwidth 8000 (kbps)
  Tunnel receive bandwidth 8000 (kbps)
  Last input never, output never, output hang never
  Last clearing of "show interface" counters 00:54:17
  Input queue: 0/375/0/0 (size/max/drops/flushes); Total output drops: 0
  Queueing strategy: fifo
  Output queue: 0/0 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
     4 packets input, 336 bytes, 0 no buffer
     Received 0 broadcasts (0 IP multicasts)
     0 runts, 0 giants, 0 throttles
     0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
     10 packets output, 1080 bytes, 0 underruns
     0 output errors, 0 collisions, 0 interface resets
     0 unknown protocol drops
     0 output buffer failures, 0 output buffers swapped out
R1#
```

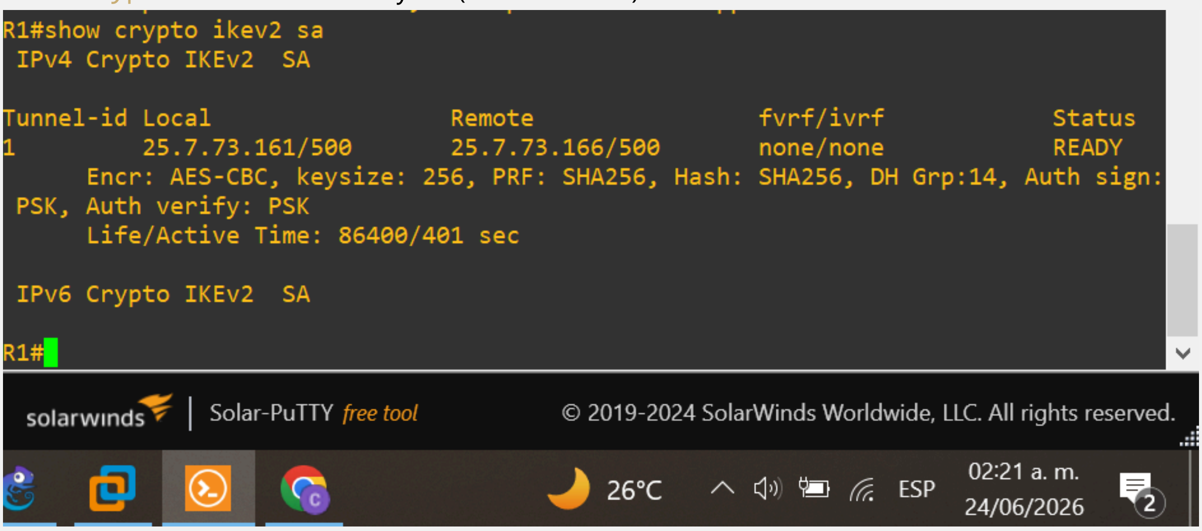
solarwinds | Solar-PuTTY *free tool*

26°C ESP 02:19 a. m. 24/06/2026

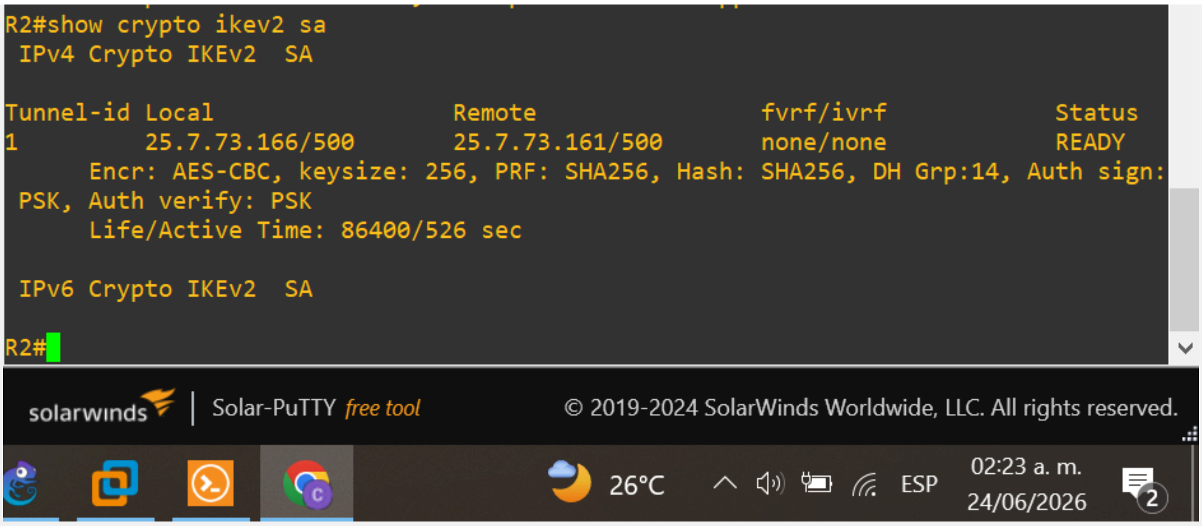


```
R2#show interface Tunnel0
Tunnel0 is up, line protocol is up
  Hardware is Tunnel
  Internet address is 25.7.73.186/30
  MTU 9976 bytes, BW 100 Kbit/sec, DLY 50000 usec,
     reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation TUNNEL, loopback not set
  Keepalive not set
  Tunnel linestate evaluation up
  Tunnel source 25.7.73.166 (GigabitEthernet2), destination 25.7.73.161
   Tunnel Subblocks:
      src-track:
         Tunnel0 source tracking subblock associated with GigabitEthernet2
          Set of tunnels with source GigabitEthernet2, 1 member (includes iterators
), on interface <OK>
  Tunnel protocol/transport GRE/IP
    Key disabled, sequencing disabled
    Checksumming of packets disabled
  Tunnel TTL 255, Fast tunneling enabled
  Tunnel transport MTU 1476 bytes
  Tunnel transmit bandwidth 8000 (kbps)
  Tunnel receive bandwidth 8000 (kbps)
  Last input never, output never, output hang never
  Last clearing of "show interface" counters 00:18:55
  Input queue: 0/375/0/0 (size/max/drops/flushes); Total output drops: 0
  Queueing strategy: fifo
  Output queue: 0/0 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
     5 packets input, 420 bytes, 0 no buffer
     Received 0 broadcasts (0 IP multicasts)
     0 runts, 0 giants, 0 throttles
     0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
     4 packets output, 432 bytes, 0 underruns
     0 output errors, 0 collisions, 0 interface resets
     0 unknown protocol drops
     0 output buffer failures, 0 output buffers swapped out
R2#s
```


3. `show crypto ikev2 sa` en R1 y R2 (Status READY)

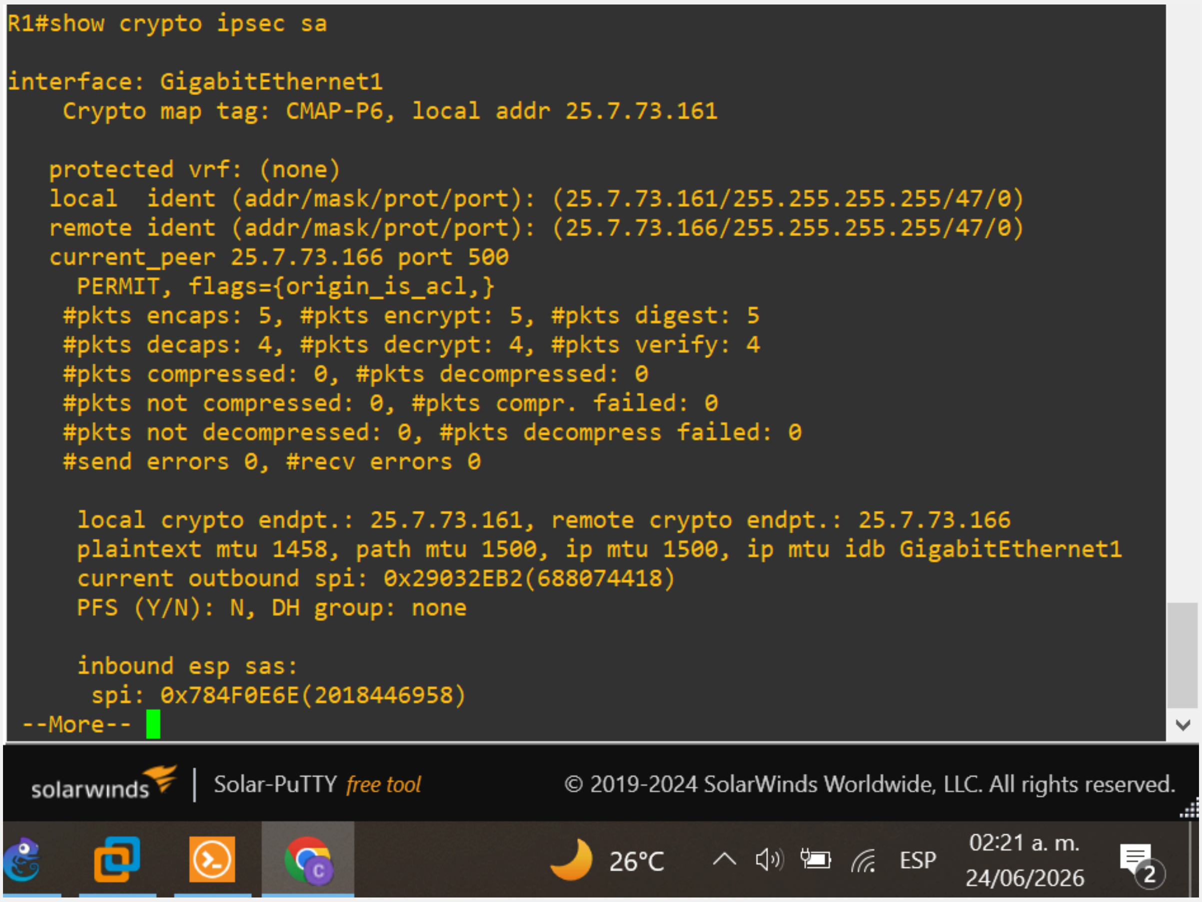


```
R1#show crypto ikev2 sa
 IPv4 Crypto IKEv2  SA

Tunnel-id Local                 Remote                fvrf/ivrf            Status
1         25.7.73.161/500       25.7.73.166/500       none/none            READY
      Encr: AES-CBC, keysize: 256, PRF: SHA256, Hash: SHA256, DH Grp:14, Auth sign:
 PSK, Auth verify: PSK
      Life/Active Time: 86400/401 sec

 IPv6 Crypto IKEv2  SA

R1#
```



```
R2#show crypto ikev2 sa
 IPv4 Crypto IKEv2  SA

Tunnel-id Local                 Remote                fvrf/ivrf            Status
1         25.7.73.166/500       25.7.73.161/500       none/none            READY
      Encr: AES-CBC, keysize: 256, PRF: SHA256, Hash: SHA256, DH Grp:14, Auth sign:
 PSK, Auth verify: PSK
      Life/Active Time: 86400/526 sec

 IPv6 Crypto IKEv2  SA

R2#
```


4. `show crypto ipsec sa` en R1 y R2 (modo Transport, protocolo 47, Status ACTIVE)

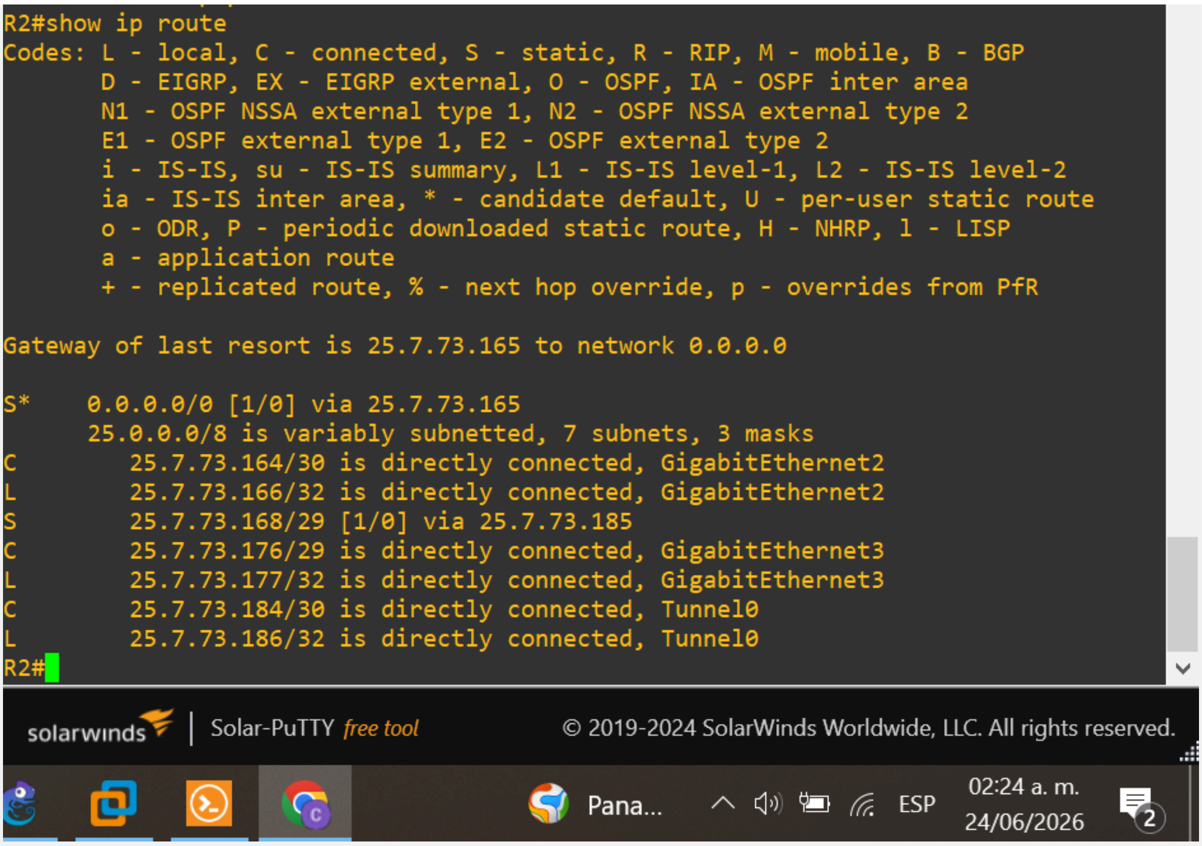


```
R1#show crypto ipsec sa

interface: GigabitEthernet1
    Crypto map tag: CMAP-P6, local addr 25.7.73.161

   protected vrf: (none)
   local  ident (addr/mask/prot/port): (25.7.73.161/255.255.255.255/47/0)
   remote ident (addr/mask/prot/port): (25.7.73.166/255.255.255.255/47/0)
   current_peer 25.7.73.166 port 500
     PERMIT, flags={origin_is_acl,}
    #pkts encaps: 5, #pkts encrypt: 5, #pkts digest: 5
    #pkts decaps: 4, #pkts decrypt: 4, #pkts verify: 4
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0

     local crypto endpt.: 25.7.73.161, remote crypto endpt.: 25.7.73.166
     plaintext mtu 1458, path mtu 1500, ip mtu 1500, ip mtu idb GigabitEthernet1
     current outbound spi: 0x29032EB2(688074418)
     PFS (Y/N): N, DH group: none

     inbound esp sas:
      spi: 0x784F0E6E(2018446958)
 --More--
```

```
R2#show crypto ipsec sa

interface: GigabitEthernet2
    Crypto map tag: CMAP-P6, local addr 25.7.73.166

   protected vrf: (none)
   local  ident (addr/mask/prot/port): (25.7.73.166/255.255.255.255/47/0)
   remote ident (addr/mask/prot/port): (25.7.73.161/255.255.255.255/47/0)
   current_peer 25.7.73.161 port 500
     PERMIT, flags={origin_is_acl,}
    #pkts encaps: 4, #pkts encrypt: 4, #pkts digest: 4
    #pkts decaps: 5, #pkts decrypt: 5, #pkts verify: 5
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0

     local crypto endpt.: 25.7.73.166, remote crypto endpt.: 25.7.73.161
     plaintext mtu 1458, path mtu 1500, ip mtu 1500, ip mtu idb GigabitEthernet2
     current outbound spi: 0x784F0E6E(2018446958)
     PFS (Y/N): N, DH group: none

     inbound esp sas:
      spi: 0x29032EB2(688074418)
 --More--
```

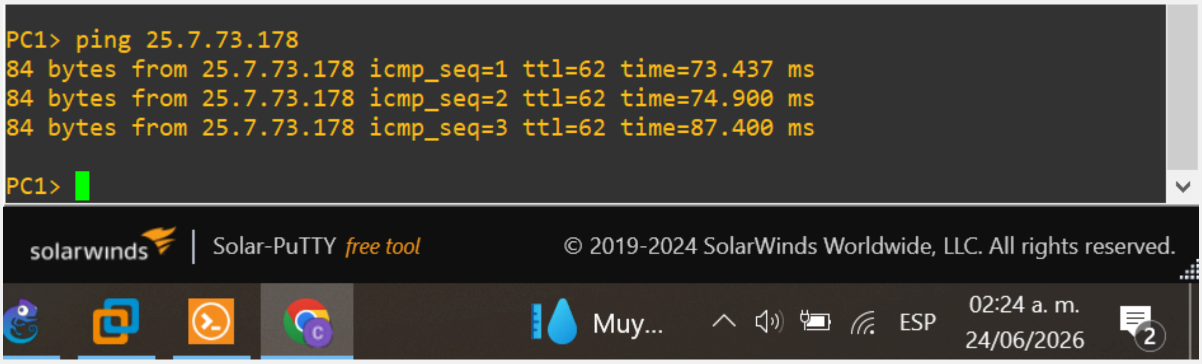
5. `show ip route` en R1 y R2 (ruta via Tunnel0)

```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is 25.7.73.162 to network 0.0.0.0

S*    0.0.0.0/0 [1/0] via 25.7.73.162
      25.0.0.0/8 is variably subnetted, 7 subnets, 3 masks
C        25.7.73.160/30 is directly connected, GigabitEthernet1
L        25.7.73.161/32 is directly connected, GigabitEthernet1
C        25.7.73.168/29 is directly connected, GigabitEthernet3
L        25.7.73.169/32 is directly connected, GigabitEthernet3
S        25.7.73.176/29 [1/0] via 25.7.73.186
C        25.7.73.184/30 is directly connected, Tunnel0
L        25.7.73.185/32 is directly connected, Tunnel0
R1#
```



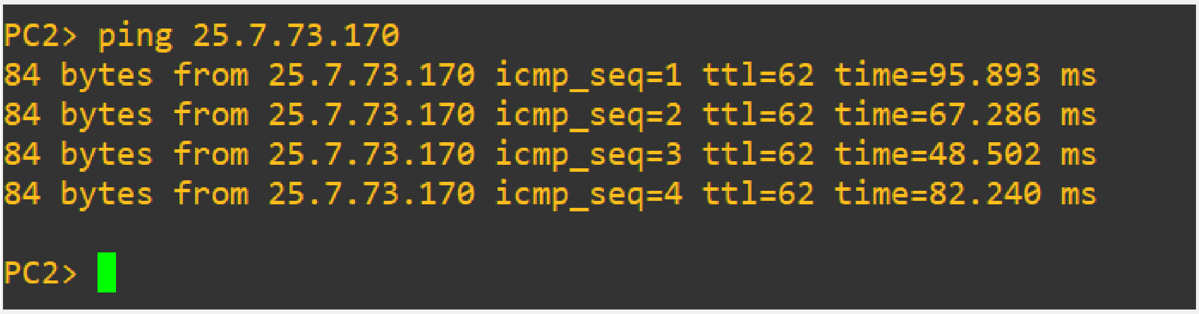
```
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is 25.7.73.165 to network 0.0.0.0

S*    0.0.0.0/0 [1/0] via 25.7.73.165
      25.0.0.0/8 is variably subnetted, 7 subnets, 3 masks
C        25.7.73.164/30 is directly connected, GigabitEthernet2
L        25.7.73.166/32 is directly connected, GigabitEthernet2
S        25.7.73.168/29 [1/0] via 25.7.73.185
C        25.7.73.176/29 is directly connected, GigabitEthernet3
L        25.7.73.177/32 is directly connected, GigabitEthernet3
C        25.7.73.184/30 is directly connected, Tunnel0
L        25.7.73.186/32 is directly connected, Tunnel0
R2#
```


6. Ping PC1 → PC2 (primer timeout y luego exitoso)



```
PC1> ping 25.7.73.178
84 bytes from 25.7.73.178 icmp_seq=1 ttl=62 time=73.437 ms
84 bytes from 25.7.73.178 icmp_seq=2 ttl=62 time=74.900 ms
84 bytes from 25.7.73.178 icmp_seq=3 ttl=62 time=87.400 ms

PC1>
```


7. Ping PC2 → PC1



```
PC2> ping 25.7.73.170
84 bytes from 25.7.73.170 icmp_seq=1 ttl=62 time=95.893 ms
84 bytes from 25.7.73.170 icmp_seq=2 ttl=62 time=67.286 ms
84 bytes from 25.7.73.170 icmp_seq=3 ttl=62 time=48.502 ms
84 bytes from 25.7.73.170 icmp_seq=4 ttl=62 time=82.240 ms

PC2>
```

Conclusión

Se configuró exitosamente una VPN IPSec IKEv2 Site-to-Site con túnel GRE (GRE over IPSec) entre R1 y R2. La combinación GRE + IPSec IKEv2 en modo Transport representa la implementación más moderna de esta arquitectura: GRE encapsula el tráfico entre las LANs (incluyendo soporte nativo de multicast/broadcast y protocolos de enrutamiento dinámico), mientras que IKEv2 negocia las Security Associations de forma más eficiente que IKEv1 (4 mensajes vs 6-9), y el modo Transport de IPSec cifra el payload GRE sin re-encapsularlo.

La SA IKEv2 alcanzó el estado READY con AES-CBC-256, SHA-256 y DH Group 14, y la conectividad entre LAN-A (25.7.73.168/29) y LAN-B (25.7.73.176/29) fue verificada exitosamente mediante ping bidireccional.